

## EDEN Finance Dashboard-Conception

### Project type and goal

- Full-stack finance analytics dashboard for EDEN Bioworks
- Shows income and expenses as an interactive table and as charts
- Users are the founders, tracking reagent sales and where money goes

### Key features

- Transaction table with sorting, filtering, pagination and search
- Pie chart of expenses by category
- Bar chart of monthly income against expenses
- Line chart of spending over time
- REST API serving JSON, AJAX filtering without page reload, responsive layout
- Date range filter on both views

### Tech stack and reasons

- Node.js with Express: one language on both sides; Flask would mean two ecosystems
- PostgreSQL: every chart is an aggregation SQL answers with GROUP BY and SUM, and the data is relational
- Bootstrap 5, DataTables.js, Chart.js
- No React: these libraries write to the DOM directly, so it would add a build step and a compatibility layer without simplifying a two-page interface

### Scope

- Prototype on sample data, no accounting integration, no login

### Repository

- [github.com/OykuBalaban/eden-finance-dashboard](https://github.com/OykuBalaban/eden-finance-dashboard)

#### Transaction table page

| EDEN Finance Dashboard                      |             |          | Transactions   Charts |     |        |
|---------------------------------------------|-------------|----------|-----------------------|-----|--------|
| [Category ▼] [From date] [To date] [Search] |             |          |                       |     |        |
| Date                                        | Description | Category | Product               | Qty | Amount |
|                                             |             |          |                       |     |        |
| ◀ 1 2 3 ▶                                   |             |          |                       |     |        |

#### Visualizer page

| EDEN Finance Dashboard             |  | Transactions   Charts                     |  |
|------------------------------------|--|-------------------------------------------|--|
| [Category ▼] [From date] [To date] |  |                                           |  |
| Pie chart<br>Expenses by category  |  | Bar chart<br>Income vs expenses per month |  |
| Line chart<br>Spending over time   |  |                                           |  |

Responsive layout using the Bootstrap 5 grid

